

Metagenomic analysis can more truly reflect the microbial composition and interaction in the sample, and study the metabolic pathway and gene function at the molecular level^[1-4].

1. Sequencing results pretreatment

(1) The original sequencing reads are dumbbell shaped structural sequences containing two end connectors, which are called polymerase reads. After the original data is interrupted from the connector and the connector sequence is filtered, the subreads are obtained. The subreads are filtered according to the standard filtering subreads by minimum length = 50.

(2) Based on subreads, CCS software is adopted (<https://github.com/PacificBiosciences/ccs>) to generate high-precision hifi reads^[5]. Here, the standard we set is min passes = 3, min RQ = 0.99, that is, all reads quality values are above Q20. Generally speaking, this part of data is the effective input data of downstream analysis.

2. Metagenome Assembly

Starting from the clean data after quality control of each sample, hifiasm-meta^[6,7] was used for genome assembly, the parameters are as follows: -t32. For contig with length ≥ 1 m, drep is used to remove redundancy^[8], and then checkm software is used (<https://ecogenomics.github.io/CheckM/>) for genome integrity assessment^[9].

3. Genome classification analysis

The genome taxonomy database gtdb (genome taxonomy database) is a standard process for constructing genome taxonomy research based on the phylogenetic analysis of a large number of genomes, so as to classify microorganisms^[10,11]. The database homepage: <http://gtdb.ecogenomic.org/>. We used gtdb to classify the assembled sequences at the genomic level, the parameters are as follows: gtdbtk classify_wf --cpus 1.

4. Genome Component prediction

Genome component prediction included the prediction of the coding gene, repetitive sequences, non-coding RNA, genomics islands, transposon, prophage, and clustered regularly interspaced short palindromic repeat sequences (CRISPR). The available steps were proceeded as follows:

1) We used the GeneMarkS^[12] program to retrieve the related coding gene.

2) The interspersed repetitive sequences were predicted using the RepeatMasker^[13] (<http://www.repeatmasker.org/>). The tandem Repeats were analyzed by the TRF^[14] (Tandem repeats finder).

3) Transfer RNA (tRNA) genes were predicted by the^[15] tRNAscan-SE. Ribosome RNA (rRNA) genes were analyzed by the rRNAmmer. Small nuclear RNAs (snRNA) were predicted by BLAST against the Rfam^[16] database.

4) The IslandPath-DIOMB^[17] program was used to predict the Genomics Islands, and transposonPSI was used to predict the transposons based on the homologous blast method. The PHAST^[18] was used for the prophage prediction (<http://phast.wishartlab.com/>) and the CRISPRFinder^[19] was used for the CRISPR identification.

5. Gene function

We used seven databases to predict gene functions. They were respective GO^[20] (Gene Ontology), KEGG^{[21][22]} (Kyoto Encyclopedia of Genes and Genomes), COG^[23] (Clusters of Orthologous Groups), NR^[24] (Non-Redundant Protein Database), TCDB^[25] (Transporter Classification Database), and, Swiss-Prot^[26]. A whole genome Blast search (E-value less than 1e-5, minimal alignment length percentage larger than 40%) was performed against above seven databases. The secretory proteins were predicted by the SignalP^[27] database, and the prediction of Type I-VII proteins secreted by the pathogenic bacteria were based on the EffectiveT3^[28] software. Meanwhile, we analyzed the secondary metabolism gene clusters by the antiSMASH^[29]. For pathogenic bacteria, we added the pathogenicity and drug resistance analyses. We used the PHI^[30] (Pathogen Host Interactions), VFDB^[31] (Virulence Factors of Pathogenic Bacteria), ARDB^[32] (Antibiotic Resistance Genes Database) to perform the above analyses. Carbohydrate-Active enzymes were predicted by the Carbohydrate-Active enZymes Database^[33].

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